Lecture 20: Singular Value Decomposition

Balls, Gaussian Clouds, and Real Orthogonal Axes

A Statistical Question

Today's Question

In statistics, a standard normal random vector is **round**:

$$x \sim N(0, I).$$

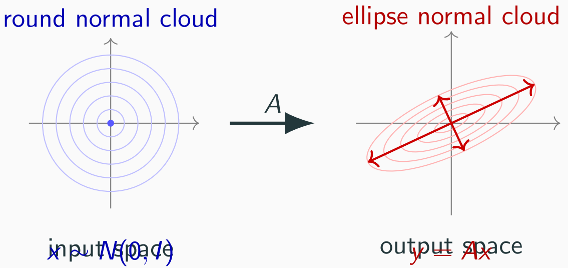
Every direction has the same variance.

Now apply a real matrix:

$$y = Ax.$$

What shape is the distribution of y?

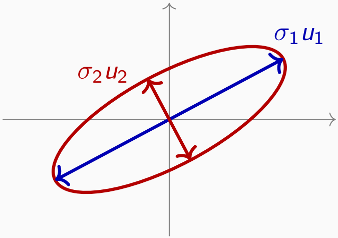
Round Cloud Becomes Ellipse Cloud



SVD asks for the axes and axis lengths of this ellipse.

What Information Do We Need?

The output cloud has two visible pieces of geometry:



- **Directions**: where are the axes? $u_1, u_2, \ldots$
- **Lengths**: how long are the axes? $\sigma_1, \sigma_2, \ldots$

Covariance Remembers the Shape

For a random vector, covariance records the shape of the cloud.

If $y = Ax$, then:

$$\mathrm{Cov}(y) = \mathrm{Cov}(Ax) = A\,\mathrm{Cov}(x)\,A^T.$$

Linear transformations change covariance by $C \mapsto ACA^T$.

Standard Normal Input

For a standard normal input:

$$\mathrm{Cov}(x) = I.$$

Therefore:

$$\mathrm{Cov}(y) = AIA^T.$$

So:

$$\boxed{\mathrm{Cov}(y) = AA^T.}$$

The First Symmetric Matrix

The output cloud is controlled by the real symmetric matrix

$$AA^T, \qquad (AA^T)^T = AA^T.$$

So the previous lecture applies:

$$AA^T = \underbrace{U}_{\text{orthogonal axes}} \underbrace{\Sigma^2}_{\text{variances along axes}} \underbrace{U^T}_{\text{measure in those axes}}.$$

The output axes come from real symmetric diagonalization.

No Unitary Matrix Is Needed Here

This lecture is an applied real-number story.

The visible objects are:

$$A^T, \qquad AA^T, \qquad A^TA, \qquad U^TU = I, \qquad V^TV = I.$$

The final real formula will be:

$$\boxed{A = U\Sigma V^T.}$$

Complex notation is only a later remark: replace T by H.

Read the Formula in Three Steps

The formula

$$A = U\Sigma V^T$$

is not just symbols. It is a three-step geometric story.

Step	Matrix	Geometric action
1	V^T	rotate the input axes
2	Σ	stretch by axis lengths
3	U	rotate to the output axes

Σ stores lengths. U stores output directions. V stores input directions.

The Same Question Without Probability

Forget Probability for One Minute

The same shape question can be asked without random variables.

Look at the unit ball:

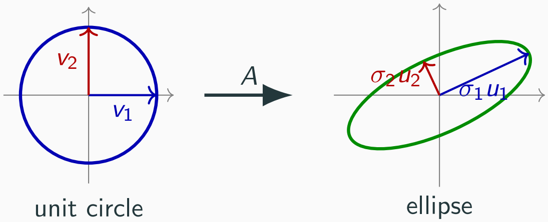
$$B_1 = \{x : \|x\| \leq 1\}.$$

Apply a matrix:

$$A(B_1) = \{Ax : \|x\| \leq 1\}.$$

What shape is $A(B_1)$?

Circle to Ellipse



The image of a round object has special axes.

Axis Equation

The special input directions are:

$$v_1,\ v_2,\ \ldots$$

They land exactly on the output axes:

$$Av_1 = \sigma_1 u_1, \qquad Av_2 = \sigma_2 u_2, \qquad \cdots$$

input axis	**stretch**	**output axis**
v_1	σ_1	u_1
v_2	σ_2	u_2
v_3	0	invisible

Geometric Definition

Definition (Singular Values)

The singular values of a real matrix A are the axis lengths of the ellipsoid

$$A(B_1) = \{Ax : \|x\| \leq 1\}.$$

They are written:

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq 0.$$

A singular value is a length, so it is never negative.

What We Must Find

SVD asks for three pieces of data:

$$\boxed{Av_i = \sigma_i u_i}$$

Symbol	Meaning	Where it lives
v_i	input axis direction	domain of A
σ_i	axis length	nonnegative number
u_i	output axis direction	codomain of A

Once we know these, the matrix itself becomes easy to describe.

Stack the Axis Equations

Write all axis equations at once:

$$A \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & 0 \end{pmatrix}}_{\Sigma}.$$

Short version:

$$\boxed{AV = U\Sigma.}$$

From Axis Equations to SVD

The input axes are orthonormal, so:

$$V^T V = I, \qquad V^{-1} = V^T.$$

Starting from:

$$AV = U\Sigma,$$

multiply on the right by V^T:

$$AVV^T = U\Sigma V^T.$$

Therefore:

$$\boxed{A = U\Sigma V^T.}$$

The Three Actions

The formula

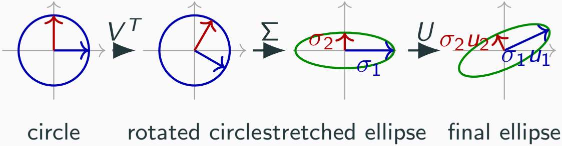
$$A = \underbrace{U}_{\text{output axes}} \underbrace{\Sigma}_{\text{axis lengths}} \underbrace{V^T}_{\text{input coordinates}}$$

means:

$$\boxed{\text{input vector}} \xrightarrow{V^T} \boxed{\text{coordinates in } V} \xrightarrow{\Sigma} \boxed{\text{stretch by } \Sigma} \xrightarrow{U} \boxed{\text{output vector}}$$

Measure, stretch, rebuild.

Circle, Rotate, Stretch, Rotate



The formula is visual: circle $\to$ circle $\to$ ellipse $\to$ rotated ellipse.

Finding the Input Axes

Where Do the Input Axes Come From?

We need directions v_i such that:

$$Av_i = \sigma_i u_i.$$

But A may be rectangular:

$$A : \mathbb{R}^n \longrightarrow \mathbb{R}^m.$$

So A itself may not have eigenvectors.

Instead, measure how much A stretches each input direction.

The Stretch-Measuring Matrix

For any input vector x:

$$\|Ax\|^2 = (Ax)^T(Ax).$$

Move the transpose:

$$(Ax)^T = x^T A^T.$$

Therefore:

$$\boxed{\|Ax\|^2 = x^T A^T Ax.}$$

So define:

$$\boxed{G = A^T A.}$$

Why $G = A^T A$ Is Symmetric

Compute:

$$G^T = (A^T A)^T.$$

Use the transpose product rule:

$$(A^T A)^T = A^T (A^T)^T.$$

Since $(A^T)^T = A$:

$$G^T = A^T A = G.$$

$A^T A$ **is a real symmetric matrix.**

The Squared Length Is Nonnegative

For every vector x:

$$x^T G x = x^T A^T A x.$$

But:

$$x^T A^T A x = (Ax)^T (Ax).$$

So:

$$x^T G x = \|Ax\|^2 \geq 0.$$

We only need this fact: $x^T A^T A x$ is a squared length.

Now Use the Previous Lecture

We proved:

Theorem (Real Symmetric Spectral Theorem)

If G is real symmetric, then it has an orthonormal eigenbasis.
Equivalently,

$$G = V\Lambda V^T$$

for an orthogonal matrix V and a real diagonal matrix Λ.

Here:

$$G = A^T A.$$

Apply It to $A^T A$

Since $A^T A$ is real symmetric:

$$A^T A v_i = \lambda_i v_i$$

with orthonormal eigenvectors:

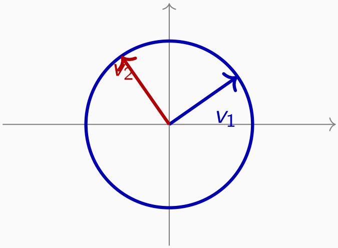
$$v_i^T v_j = \delta_{ij}.$$

Stack them:

$$V = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}, \qquad V^T V = I.$$

These eigenvectors are the input axes of the ellipsoid.

Input Axes Still Live on the Circle



input unit circle

v_i are not output axes yet. They are special directions on the input circle.

Why Eigenvalues Are Nonnegative

Suppose:

$$A^T A v_i = \lambda_i v_i, \qquad \|v_i\| = 1.$$

Then:

$$\lambda_i = \lambda_i v_i^T v_i = v_i^T (A^T A v_i) = v_i^T A^T A v_i.$$

But:

$$v_i^T A^T A v_i = \|A v_i\|^2 \geq 0.$$

So:

$$\boxed{\lambda_i \geq 0.}$$

Singular Values Are Square Roots

The eigenvalues of $A^T A$ are nonnegative:

$$\lambda_i \geq 0.$$

So we may define:

$$\boxed{\sigma_i = \sqrt{\lambda_i}.}$$

Then:

$$A^T A v_i = \sigma_i^2 v_i.$$

The square root appears because $A^T A$ measures squared length.

Check the Length

Assume $A^T A v_i = \sigma_i^2 v_i$ and $\|v_i\| = 1$.

Then:

$$\begin{aligned} \|Av_i\|^2 &= v_i^T A^T A v_i \\ &= v_i^T (\sigma_i^2 v_i) \\ &= \sigma_i^2 v_i^T v_i = \sigma_i^2. \end{aligned}$$

Therefore:

$$\boxed{\|Av_i\| = \sigma_i.}$$

Output Axes

Now Build the Output Axes

We already know:

$$A^T A v_i = \sigma_i^2 v_i, \qquad \|Av_i\| = \sigma_i.$$

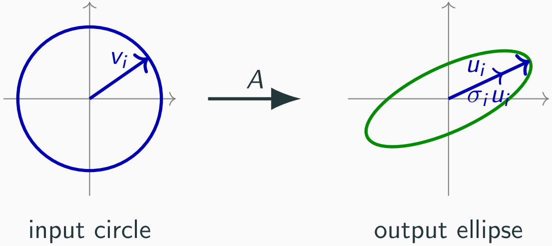
So if $\sigma_i > 0$, define:

$$\boxed{u_i = \frac{Av_i}{\sigma_i}.}$$

Then the axis equation becomes:

$$\boxed{Av_i = \sigma_i u_i.}$$

Input Axis Lands on Output Axis



v_i is an input direction. u_i is the normalized output direction.

Check u_i Has Length 1

By definition:

$$u_i = \frac{Av_i}{\sigma_i}.$$

So:

$$\|u_i\| = \left\| \frac{Av_i}{\sigma_i} \right\| = \frac{\|Av_i\|}{\sigma_i} = \frac{\sigma_i}{\sigma_i} = 1.$$

Each nonzero output axis direction is a unit vector.

Why Different Output Axes Are Orthogonal

For $i \neq j$:

$$u_i^T u_j = \frac{(Av_i)^T (Av_j)}{\sigma_i \sigma_j}.$$

Move A to the middle:

$$u_i^T u_j = \frac{v_i^T A^T A v_j}{\sigma_i \sigma_j}.$$

Use $A^T A v_j = \sigma_j^2 v_j$:

$$u_i^T u_j = \frac{\sigma_j^2}{\sigma_i \sigma_j} v_i^T v_j = 0.$$

Inner-Product Table

The output vectors are orthonormal:

	u_1	u_2	u_3
u_1^T	1	0	0
u_2^T	0	1	0
u_3^T	0	0	1

Matrix form:

$$U^T U = I.$$

What If $\sigma_i = 0$?

If $\sigma_i = 0$, then:

$$A^T A v_i = 0.$$

Compute the output length:

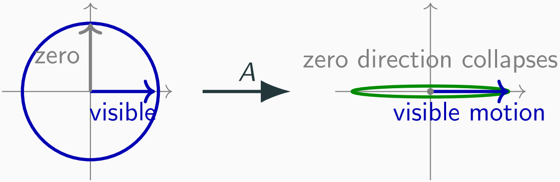
$$\|Av_i\|^2 = v_i^T A^T A v_i = 0.$$

Therefore:

$$\boxed{Av_i = 0.}$$

Zero singular value = invisible input direction.

Visible and Invisible Directions



The null space is exactly the collection of invisible directions.

Complete the Output Basis

The vectors coming from nonzero singular values span Col(A).

If the output space is larger, complete them:

$$u_1, \ldots, u_r \quad \rightsquigarrow \quad u_1, \ldots, u_r, u_{r+1}, \ldots, u_m.$$

The extra vectors live perpendicular to the column space:

$$\mathrm{Col}(A)^{\perp} = \mathrm{Null}(A^T).$$

This gives a full orthogonal matrix U.

Stack Everything

For every input axis:

$$Av_i = \sigma_i u_i.$$

Stack columns:

$$A \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\Sigma}_{\text{rectangular diagonal}}.$$

So:

$$\boxed{AV = U\Sigma.}$$

Singular Value Decomposition

Since V is orthogonal:

$$V^{-1} = V^T.$$

From:

$$AV = U\Sigma,$$

multiply by V^T:

$$AVV^T = U\Sigma V^T.$$

Therefore:

$$\boxed{A = U\Sigma V^T.}$$

Theorem

Theorem (Singular Value Decomposition)

For every real $m \times n$ matrix A, there exist orthogonal matrices U and V, and a rectangular diagonal matrix Σ with nonnegative entries, such that

$$A = U\Sigma V^T.$$

Every real matrix maps balls to ellipsoids with orthogonal axes.

Proof Photograph

A^TA	is real symmetric	previous lecture applies
$A^TAv_i = \sigma_i^2 v_i$	gives	input axes v_i
$\lVert Av_i \rVert = \sigma_i$	gives	axis lengths σ_i
$u_i = Av_i/\sigma_i$	gives	output axes u_i
$Av_i = \sigma_i u_i$	stack	$AV = U\Sigma$
$V^{-1} = V^T$	gives	$A = U\Sigma V^T$

The whole proof is real symmetric diagonalization of A^TA.

Method Summary

The Method

To decompose a real matrix A:

1. Compute the real symmetric matrix $A^T A$.
2. Orthogonally diagonalize it:

$$A^T A = V\Sigma^2 V^T.$$

3. For each nonzero σ_i, define:

$$u_i = \frac{Av_i}{\sigma_i}.$$

4. Complete the u_i's to an orthonormal basis if needed.
5. Stack the equations $Av_i = \sigma_i u_i$.

What Each Piece Means

The decomposition is:

$$\boxed{A = U\Sigma V^T.}$$

Piece	Meaning
V	orthonormal input axes on the unit circle
Σ	axis lengths of the output ellipsoid
U	orthonormal output axes

Input axes $\rightarrow$ stretch lengths $\rightarrow$ output axes.

The Whole Story in One Line

For every input axis:

$$Av_i = \sigma_i u_i.$$

Stack all columns:

$$A \underbrace{\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}}_{V} = \underbrace{\begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{pmatrix}}_{U} \underbrace{\Sigma}_{\text{axis lengths}} .$$

Therefore:

$$\boxed{A = U\Sigma V^T.}$$

Final Summary

SVD answers one geometric question:

What does A do to a round ball?

Answer:

$$\boxed{\text{round ball} \xrightarrow{A} \text{ellipsoid with orthogonal axes.}}$$

The axis data is exactly:

$$\boxed{A = U\Sigma V^T.}$$